

To give you an indication of how ugly things can get with leverage, Table 4.4 provides the probability of losing a quarter of your initial investment from a leveraged transaction. The question is, “What are the odds that if you start off with \$10,000 and borrow, you’ll be left with less than \$7,500 after 12 months?” This would be a 25% loss of equity capital.

TABLE 4.4 Leverage: The Ugly—What Are the Chances of Losing a Quarter of Your Initial Investment? (Market Expected Return: 15%; Volatility: 30%)

Leverage Ratio	Interest Cost		
	3%	5%	7%
0%	9.1%	9.1%	9.1%
25%	12.6%	12.9%	13.1%
50%	15.3%	15.9%	16.4%
100%	19.3%	20.2%	21.2%
250%	25.2%	26.8%	28.4%

Source: Moshe Milevsky and the IFID Centre, 2008.

If you don’t leverage at all, the chances of suffering such a drastic reduction of capital are about 9%. But if you choose the 100% leverage route, and you pay a margin interest cost of 5%, the odds of losing a quarter of your initial investment are close to 20%. That’s a one-in-five chance. Why are the odds so high? After all, the odds of doing very well were quite high as well. How can the odds be so high for both the good and the ugly?

Well, that’s the key point with investment leverage. You are subjecting yourself to extremes. Things can go very well—or very badly. Very well if the market goes up; very badly if the market goes down. I like to say that leverage exaggerates both the severe upside and the severe downside.

On a technical level, you have increased the potential return of your investment, but you have increased the uncertainty as well. The higher uncertainty means that you are more likely to experience the extreme cases. In the good times, the extremes will be welcome. In the bad, they can bring you to the brink of bankruptcy.